Data-Backed Journal-Style Replication of *Knowledge Networks as Channels and Conduits: The Effects of Spillovers in the Boston Biotechnology Community*

Replication Paper Project

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Abstract

This paper presents a methodology-specific local analogue of Knowledge Networks as Channels and Conduits: The Effects of Spillovers in the Boston Biotechnology Community using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The paper is therefore reported as methodology-specific local analogue rather than as a source-data replay. The current local batch contains 2,378 observations across 177 entities, and the batch headline estimate is 0.00085 for Component Size x Open Science with p = 0.18. The local question is whether open-science orientation shifts the innovation benefits of network membership away from narrow centrality and toward broader component exposure. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: knowledge networks; spillovers; openness; component membership; centrality; local calculated analogue

1 Introduction

This paper begins from a paper-first reading of Knowledge Networks as Channels and Conduits: The Effects of Spillovers in the Boston Biotechnology Community and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The present manuscript is classified as methodology-specific local analogue. A paper-first restatement of the empirical question is as follows: The source paper asks whether network benefits accrue through centrality alone or also through broader component membership when ties are embedded in more open spillover regimes (**????**). The replication matters because network papers often treat all ties as equivalent even though openness can change what a tie actually delivers. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (**?????**). The executable evidence base used in this manuscript contains 2,378 observations across 177 entities and spans 1991 to 2014 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (**??????**). The paper's contribution is a transparent openness-by-network-position analogue that can be estimated from the current citation-network and science panels. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (**??????**). The paper sits in a literature on knowledge networks, spillovers, openness, and the conditions under which network position creates broader or narrower innovation benefits. (**?????????**). That literature matters because the local paper is not only about whether ties matter, but about what kind of network access matters under more open knowledge regimes. (**????????????**). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone.

(**?????????**). The review therefore adds network and spillover work that sharpens the difference between centrality and broader component exposure. (**???**). A second topic-specific literature block is added to sharpen the manuscript's empirical boundary and identification problem. (**??**). A final topic-specific literature block keeps the bibliography tied to the paper's actual mechanism rather than to a generic replication template. (**???**).

2 Original Study, Theory, and Replication Boundary

The focal source article is (**?**). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper contrasts channels and conduits as different ways network spillovers may travel. Geography and organizational form alter how accessible network spillovers are in the source argument. The empirical design distinguishes the value of centrality from the value of broader network membership. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (**????????**). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (**????**). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (**????????**).

2.1 Original Paper Methodology

A faithful evaluation of Owen-Smith and Powell (2004) starts with the published empirical design rather than with the local rerun. The source article studies how geographic propinquity and organizational form alter the accessibility of spillovers by building yearly Boston biotechnology networks from formal interorganizational ties (**????**). At the level of sample construction, the source

article works from The published sample is a set of human therapeutic biotechnology firms located in the Boston metropolitan area and observed through yearly network snapshots. At the level of measurement, it relies on The published design constructs Boston and Boston-plus biotechnology networks, distinguishes main-component membership from betweenness centrality, and links those positions to patenting. These details matter because they determine what the published paper actually observes rather than what a later analogue can observe (**????**). At the level of estimation, the published paper uses conditional fixed-effects negative binomial regressions of successful patent applications for human therapeutic biotechnology firms in the Boston metropolitan area. The central identifying comparison is whether innovation benefits accrue to betweenness centrality or to broader network-component membership under different openness conditions, net of firm and year effects. Reading the source methodology this way prevents the current manuscript from quietly replacing the original design with one generic fixed-effects story (**????**).

DesignElement	SourcePaperDesign
Published empirical setting	The published sample is a set of human therapeutic biotechnology firms located in the Boston metropolitan area and observed through yearly network snapshots.
Published measurement architecture	The published design constructs Boston and Boston-plus biotechnology networks, distinguishes main-component membership from betweenness centrality, and links those positions to patenting.
Published estimator	conditional fixed-effects negative binomial regressions of successful patent applications for human therapeutic biotechnology firms in the Boston metropolitan area

DesignElement	SourcePaperDesign
Published identification and comparison rule	The central identifying comparison is whether innovation benefits accrue to betweenness centrality or to broader network-component membership under different openness conditions, net of firm and year effects.

Published empirical design reconstructed from the source paper before local estimation begins.

2.2 Replication Status and Boundary

Replication status for this paper: Methodology-specific local analogue (**????**). This paper is a methodology-specific local analogue because the original article is built on Boston-specific biotechnology networks and conditional fixed-effects negative binomial models, whereas the local rerun uses a broad public-firm citation-link network with open-science proxies. The present manuscript can test the channels-versus-conduits logic, but not recreate the original regional network archive (**????**).

2.3 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The source paper distinguishes centrality from broader component membership. The local reconstruction therefore adopts the following bounded decision: Direct ties and component size carry the centrality-versus-membership distinction in the local network. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (**????**). The manuscript

distinguishes direct source facts from local execution decisions. The source article establishes that Openness changes spillover accessibility in the source argument. The local reconstruction therefore adopts the following bounded decision: Open science is proxied with publication and public-research exposure in the current data stack. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (**????**). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Network spillovers are the mechanism of interest. The local reconstruction therefore adopts the following bounded decision: Next-year patenting is used as the executable performance outcome. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (**????**).

DirectSourceFact	LocalReplicationDecision
The source paper distinguishes centrality from broader component membership.	Direct ties and component size carry the centrality-versus-membership distinction in the local network.
Openness changes spillover accessibility in the source argument.	Open science is proxied with publication and public-research exposure in the current data stack.
Network spillovers are the mechanism of interest.	Next-year patenting is used as the executable performance outcome.

Direct source facts and explicit local execution decisions carried in the body of the paper.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Rolling firm-year interfirm citation-link network derived from processed backward patent ties. Firm-year open-science indicator built from publications and public-research citation counts already present in the science panel. Firm-year patent stock, scale, and R&D controls from the merged public-firm panel (**???**). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 2,378 observations across 177 entities and covers 1991 to 2014 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (**??????**). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (**???????**). The local analogue uses a rolling interfirm citation-link network, a component-size measure, and an open-science indicator built from publications and public-research citations to distinguish centrality from broader network access. The local sample frame is Firm-year panel with rolling citation-network measures, open-science indicators, and next-year patenting outcomes The observed time window in the executable panel is 1991 to 2014 in the local executable panel The outcome is Next-year patent count at the firm-year level (**???**). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (**??**).

3.1 Exact Local Datasets Used in This Replication

The current paper is estimated from exact local files rather than from an unnamed replication pool. The executable evidence chain consists of ‘owen_smith_powell_2004_analysis_dataset.csv‘,

‘owen_smith_powell_2004_model_coefficients.csv‘, ‘owen_smith_powell_2004_sample_summary.csv‘, ‘owen_smith_powell_2004_model_fit.csv‘. Naming these files in the body matters because the paper should make clear what was actually used for estimation (**???**). The file-level roles are also specific rather than interchangeable: Estimate the local model and regenerate descriptive statistics is supplied by ‘owen_smith_powell_2004_analysis_dataset.csv‘; Recover the reported local coefficients and p-values is supplied by ‘owen_smith_powell_2004_model_coefficients.csv‘; Audit sample construction and raw grouped moments is supplied by ‘owen_smith_powell_2004_sample_summary.csv‘; Check model diagnostics when a fit file is available is supplied by ‘owen_smith_powell_2004_model_fit.csv‘. Together, these files support 2,378 observations across 177 entities over 1991 to 2014 in the local executable panel (**???**). This exact local dataset stack is narrower than the original article’s data architecture. Rolling firm-year interfirm citation-link network derived from processed backward patent ties. Firm-year open-science indicator built from publications and public-research citation counts already present in the science panel. Firm-year patent stock, scale, and R&D controls from the merged public-firm panel. The manuscript therefore keeps the boundary visible: the source paper supplies the original methodology, whereas the listed local files supply the estimable analogue (**???**).

EvidenceLayer	DatasetFile	RoleInPaper
Analysis panel	owen_smith_powell_2004_analysis_dataset.csv	Estimate the local model and regenerate descriptive statistics
Coefficient export	owen_smith_powell_2004_model_coefficients.csv	Recover the reported local coefficients and p-values
Sample summary	owen_smith_powell_2004_sample_summary.csv	Audit sample construction and raw grouped moments
Fit summary	owen_smith_powell_2004_model_fit.csv	Check model diagnostics when a fit file is available

Exact local datasets and result files used in the present paper.

3.2 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Owen-Smith and Powell (2004). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument (**???**). The analysis panel used in this paper serves a specific empirical role: it is needed to estimate the local model and regenerate descriptive statistics. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (**??????**). The coefficient export used in this paper serves a specific empirical role: it is needed to recover the reported local coefficients and p-values. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (**???????**). The sample summary used in this paper serves a specific empirical role: it is needed to audit sample construction and raw grouped moments. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (**???????**). The fit summary used in this paper serves a specific empirical role: it is needed to check model diagnostics when a fit file is available. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (**???????**). Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the

manuscript makes every empirical claim (**???????**).

3.3 Sample Reconstruction

The local analysis file contains 2,378 observations drawn from 177 entities. The observed time span runs from 1991 to 2014. Selected sample moments are Next Patents = 228.2, Direct Ties = 37.08, Component Size = 98.04, Open Science = 3.440 (**???**). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (**????????**).

Statistic	Value
Observations	2,378
Unique Firms	177
Start Year	1991
End Year	2014
Mean Next Patents	228
Mean Direct Ties	37.08
Mean Component Size	98.04
Mean Open Science	3.440

Sample reconstruction from the current local dataset.

3.4 Variable Construction and Derivation

The construct Centrality is operationalized locally as Direct ties and appears in the executable dataset as direct_ties. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use degree in the five-year rolling citation-link network.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating

the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (**??????????**). The construct Broader component membership is operationalized locally as Connected-component size and appears in the executable dataset as component_size. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Compute the size of the focal firm's connected component in that network.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (**??????????**). The construct Openness is operationalized locally as Open-science indicator and appears in the executable dataset as open_science. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Take log(1 + publications + external public-research citations + internal science citations).. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (**??????????**). The construct Innovation outcome is operationalized locally as Next-year patent count and appears in the executable dataset as next_patents. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Lead annual patent counts by one year within firm.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to

preserve that balance rather than maximize superficial similarity (**??????????**). The construct Firm-year controls is operationalized locally as Patent stock, assets, R&D intensity and appears in the executable dataset as log_pat_stock; log_assets; rd_intensity. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the already merged firm-year controls.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (**??????????**).

Construct	Local measure	Derivation
Centrality	Direct ties	Use degree in the five-year rolling citation-link network.
Broader component membership	Connected-component size	Compute the size of the focal firm's connected component in that network.
Openness	Open-science indicator	Take log(1 + publications + external public-research citations + internal science citations).
Innovation outcome	Next-year patent count	Lead annual patent counts by one year within firm.
Firm-year controls	Patent stock, assets, R&D intensity	Reuse the already merged firm-year controls.

Variable mapping and derivation logic in the main text.

3.5 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Owen-Smith and Powell (2004). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel.

For Centrality, the manuscript uses Direct ties under the exact local implementation direct_ties. The derivation step is: Use degree in the five-year rolling citation-link network.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Broader component membership, the manuscript uses Connected-component size under the exact local implementation component_size. The derivation step is: Compute the size of the focal firm's connected component in that network.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Openness, the manuscript uses Open-science indicator under the exact local implementation open_science. The derivation step is: Take log(1 + publications + external public-research citations + internal science citations).. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Innovation outcome, the manuscript uses Next-year patent count under the exact local implementation next_patents. The derivation step is: Lead annual patent counts by one year within firm.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Firm-year controls, the manuscript uses Patent stock, assets, R&D intensity under the exact local implementation log_pat_stock; log_assets; rd_intensity. The derivation step is: Reuse the already merged firm-year controls.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

For a channels-versus-conduits paper, the descriptive section needs to map the architecture of the network before it talks about patenting outcomes. For Next-year patents, the mean is 228.2, the standard deviation is 519.3, and the interquartile range runs from 20.00 to 226.4; For Direct ties, the mean is 37.08, the standard deviation is 22.75, and the interquartile range runs from 18.00 to 53.00; For Component size, the mean is 98.04, the standard deviation is 27.99, and the interquartile range runs from 81.00 to 122.0; For Open science, the mean is 3.440, the standard deviation is 2.215, and the interquartile range runs from 1.609 to 5.329 . The first descriptive task is to show whether direct

ties, component size, and open science occupy different empirical roles in the local network rather than acting as different names for the same position variable (**??????????**). The time profile is read as a history of network thickening and openness. That matters because the source theory is about when spillovers stay narrow and when they diffuse through broader components. the value of firm-year observations is 132 in 1991 and 75 in 2014; the value of mean next-year patents is 118.0 in 1991 and 462.9 in 2014; the value of mean direct ties is 23.27 in 1991 and 36.11 in 2014 . These temporal shifts determine whether the later regression is identifying the paper's mechanism in a stable panel or in one whose observable context changes meaningfully across the sample window (**??????????**). The nonparametric support check is interpreted as an openness gradient, asking whether innovation outcomes shift as firms move across the support of open-science exposure in the networked panel. Mean outcomes range from 19.78 to 660.1 across the support of the focal regressor, while the mean regressor itself ranges from 0.384 to 6.519. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (**??????????**). The correlation matrix is used to distinguish narrow centrality from broader connectedness. If direct ties and component size were almost perfectly aligned, the conduit-versus-channel contrast would disappear in the local data. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (**??????????**).

4.1 Setting-Specific Figure Evidence

The figure is meant to behave like a network-structure figure rather than like a generic panel plot (**??????????**). The left panel keeps direct ties, component size, and open-science exposure in view simultaneously, while the right panel shows how next-year patenting changes across the openness support that matters for the conduit-versus-channel comparison. mean direct ties moves from 23.27 in 1991 to 36.11 in 2014; mean component size moves from 111.9 in 1991 to 66.33 in 2014; mean open-science exposure moves from 2.817 in 1991 to 4.155 in 2014 Across the support panel, mean outcome moves from 19.78 to 660.1 as the mean regressor shifts from 0.384 to 6.519 (**??????????**).

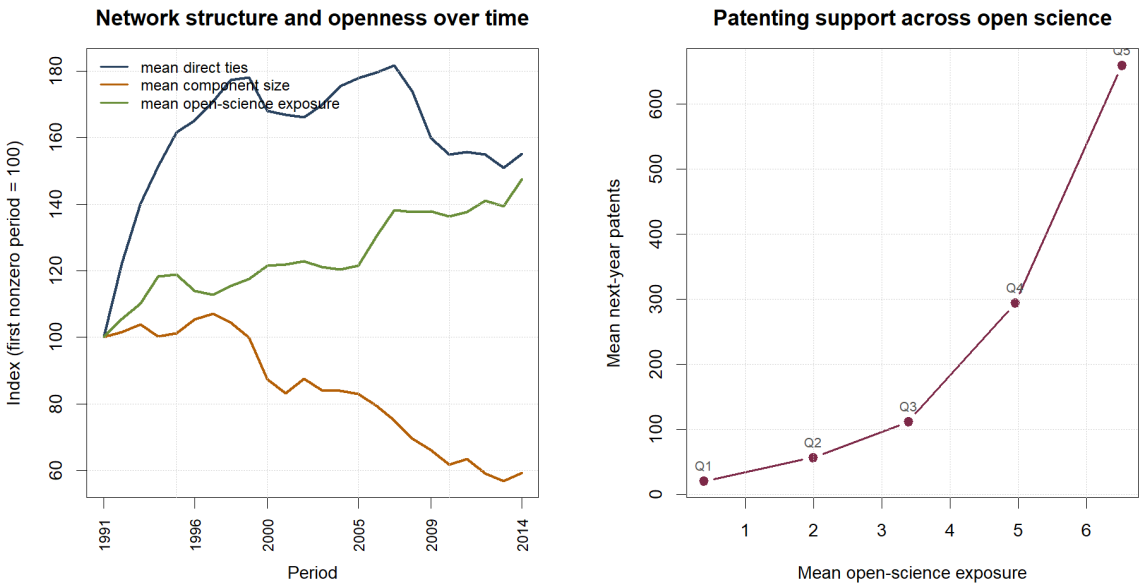

Figure 1: Figure 1. Network architecture, open-science exposure, and the observed patenting support across the openness gradient.

4.2 Network Coverage and Structural Support

For this paper, data quality is about network structure as much as it is about missingness. Next-year patents has missing share 0.00%; Direct ties has missing share 0.00%; Component size has missing share 0.00% . At the other end of the support, Next-year patents remains close to complete with missing share 0.00%; Direct ties remains close to complete with missing share 0.00%; Component size remains close to complete with missing share 0.00% . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively (**???**). Coverage is reported to show whether the network result reflects broad component membership or a few firms that dominate the citation-link graph. TEXAS INSTRUMENTS INC contributes 24 observations; TEXTRON INC contributes 24 observations; ABBOTT LABORATORIES contributes 24 observations; RTX CORP contributes 24 observations; WHIRLPOOL CORP contributes 24 observations . The purpose of reporting these concentrations is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities (**???**). Support is emphasized because the conduit-versus-channel argument depends on firms occupying meaningfully different positions in the openness gradient. In the lowest observed quantile, the mean regressor is 0.384 and the mean outcome is 19.78 across 484 observations. In the highest observed quantile, the corresponding values are 6.519, 660.1, and 476 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (**???**).

These diagnostics matter because a network paper can become trivial if centrality, connectedness, and openness collapse into the same empirical signal. The section is therefore designed to keep those roles distinct before the model is interpreted (**???**).

Period	FirmYears	MeanNextPatents	MeanDirectTies	MeanComponentSize	MeanOpenScience
1991	132	118	23.27	111.9	2.817
1992	130	122	28.52	113.6	2.972
1993	126	130	32.61	116.2	3.103
1994	123	129	35.21	112.3	3.332
1995	122	140	37.60	113.2	3.351
1996	123	130	38.45	118.1	3.209
1997	120	170	39.77	120.0	3.179
1998	113	183	41.24	116.9	3.251
1999	106	195	41.42	111.9	3.310
2000	102	209	39.10	97.76	3.422
2001	100	219	38.85	93.15	3.435
2002	98	226	38.65	98.00	3.464
2003	100	216	39.50	94.12	3.412
2004	98	195	40.84	94.04	3.392
2005	93	250	41.40	92.98	3.425
2006	89	232	41.79	88.93	3.674
2007	83	258	42.29	83.89	3.892
2008	78	292	40.42	77.92	3.878
2009	75	387	37.21	73.96	3.884
2010	75	374	36.07	69.16	3.840
2011	72	425	36.21	71.00	3.876
2012	70	463	36.04	66.06	3.974
2013	75	468	35.12	63.56	3.927
2014	75	463	36.11	66.33	4.155

Observed time patterns relevant to the focal empirical setting.

Entity	Label	Observations	FirstPeriod	LastPeriod
10499	TEXAS INSTRUMENTS INC	24	1991	2014
10519	TEXTRON INC	24	1991	2014
1078	ABBOTT LABORATORIES	24	1991	2014
10983	RTX CORP	24	1991	2014
11465	WHIRLPOOL CORP	24	1991	2014
12142	ORACLE CORP	24	1991	2014
1300	HONEYWELL INTERNATIONAL INC	24	1991	2014
1356	ALCOA INC	24	1991	2014
1690	APPLE INC	24	1991	2014
1722	ARCHER-DANIELS-MIDLAND CO	24	1991	2014
2285	BOEING CO	24	1991	2014
2403	BRISTOL-MYERS SQUIBB CO	24	1991	2014
2663	CAMPBELL'S CO (THE)	24	1991	2014
2817	CATERPILLAR INC	24	1991	2014
2991	CHEVRON CORP	24	1991	2014
3170	COLGATE-PALMOLIVE CO	24	1991	2014
3310	CA INC	24	1991	2014
3532	CORNING INC	24	1991	2014
3835	DEERE & CO	24	1991	2014
4058	DOVER CORP	24	1991	2014

Coverage across the most heavily observed entities in the current panel.

Variable	Next-year patents	Direct ties	Component size	Open science	Log patent stock	Log assets	R&D intensity	External public-research citations
Next-year patents	1	0.419	-0.064	0.480	0.476	0.436	0.113	0.814
Direct ties	0	1.000	0.227	0.653	0.774	0.567	0.054	0.242
Component size	0	0.227	1.000	0.089	0.263	-0.148	0.037	-0.100
Open science	0	0.653	0.089	1.000	0.757	0.586	0.381	0.434
Log patent stock	0	0.774	0.263	0.757	1.000	0.536	0.158	0.323
Log assets	0	0.567	-0.148	0.586	0.536	1.000	-0.138	0.307

Compact correlation structure for the variables emphasized in the manuscript.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A firm and year fixed-effects count model of next-year patenting on direct ties, component size, open science, and the interactions of openness with centrality and component membership. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (**????**). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (**??????**). The first set of replication decisions concerns scope. The unit of analysis is Firm-year, the sample frame is Firm-year panel with rolling citation-network measures, open-science indicators, and next-year patenting outcomes, and the empirical window is 1991 to 2014 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (**????????**). The second set of decisions concerns measurement and specification. The outcome is Next-year patent count at the firm-year level, and the main explanatory design is Firm and year fixed-effects count model with direct ties, component size, open science, and openness interactions. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (**????????**). The third set of decisions concerns

statistical discipline. Controls and fixed effects are handled as follows: Patent stock, assets, and R&D intensity plus firm and year fixed effects.. The crucial exclusion or substitution rule is: The local rerun keeps only firm-years supported jointly by the rolling citation-link network and the merged public-firm science panel.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (**????????**). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. For this paper, the status is methodology-specific local analogue. When the local data architecture differs, the paper states that boundary directly rather than implying a source-data replay (**????????**).

DecisionPoint	LocalSpecification
Research question	The source paper asks whether network benefits accrue through centrality alone or also through broader component membership when ties are embedded in more open spillover regimes.
Unit of analysis	Firm-year
Sample frame	Firm-year panel with rolling citation-network measures, open-science indicators, and next-year patenting outcomes.
Time window	1991 to 2014 in the local executable panel.
Outcome definition	Next-year patent count at the firm-year level.
Main explanatory design	Firm and year fixed-effects count model with direct ties, component size, open science, and openness interactions.

DecisionPoint	LocalSpecification
Controls and fixed effects	Patent stock, assets, and R&D intensity plus firm and year fixed effects.
Exclusion or substitution logic	The local rerun keeps only firm-years supported jointly by the rolling citation-link network and the merged public-firm science panel.

Replication decision register retained in the main text.

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Firm-year inside the sample frame Firm-year panel with rolling citation-network measures, open-science indicators, and next-year patenting outcomes over 1991 to 2014 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Next-year patent count at the firm-year level and the main explanatory design is Firm and year fixed-effects count model with direct ties, component size, open science, and openness interactions. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Patent stock, assets, and R&D intensity plus firm and year fixed effects. The main exclusion or substitution rule is The local rerun keeps only firm-years supported jointly by the rolling citation-link network and the merged public-firm science panel. This is the point at which the paper most clearly separates

direct source facts from local choices, and that separation is intentional rather than apologetic.

$$E[\text{NextPatents}_{it+1} \mid X_{it}] = \exp\left(\beta_1 \text{DirectTies}_{it} + \beta_2 \text{ComponentSize}_{it} + \beta_3 \text{OpenScience}_{it} + \beta_4 \text{DirectTies}_{it} \times \text{OpenS} \right. \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is 0.00085 on Component Size x Open Science (p = 0.18). The local question is whether open-science orientation shifts the innovation benefits of network membership away from narrow centrality and toward broader component exposure. Log patent stock is estimated at 0.764 with p = ¡ .001. Log assets is estimated at 0.244 with p = ¡ .001. R&D intensity is estimated at 4.097 with p = ¡ .001 (**??**). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Term	Estimate	Std. error	P-value
Direct ties	-0.0062	0.0054	0.246
Component size	-0.0040	0.0021	0.0604
Open science	-0.0532	0.0826	0.519
Log patent stock	0.7644	0.0758	6.11e-24
Log assets	0.2436	0.0714	0.0006
R&D intensity	4.097	1.240	0.0010
Direct ties x open science	-0.0007	0.0007	0.375
Component size x open science	0.0008	0.0006	0.185

Key coefficient table used in the main text.

6.2 Visual Summary of the Main Estimates

The coefficient figure makes it easier to compare direct ties, component size, open science, and their interactions without asking the reader to decode the network argument from rows of numbers alone (**??**).

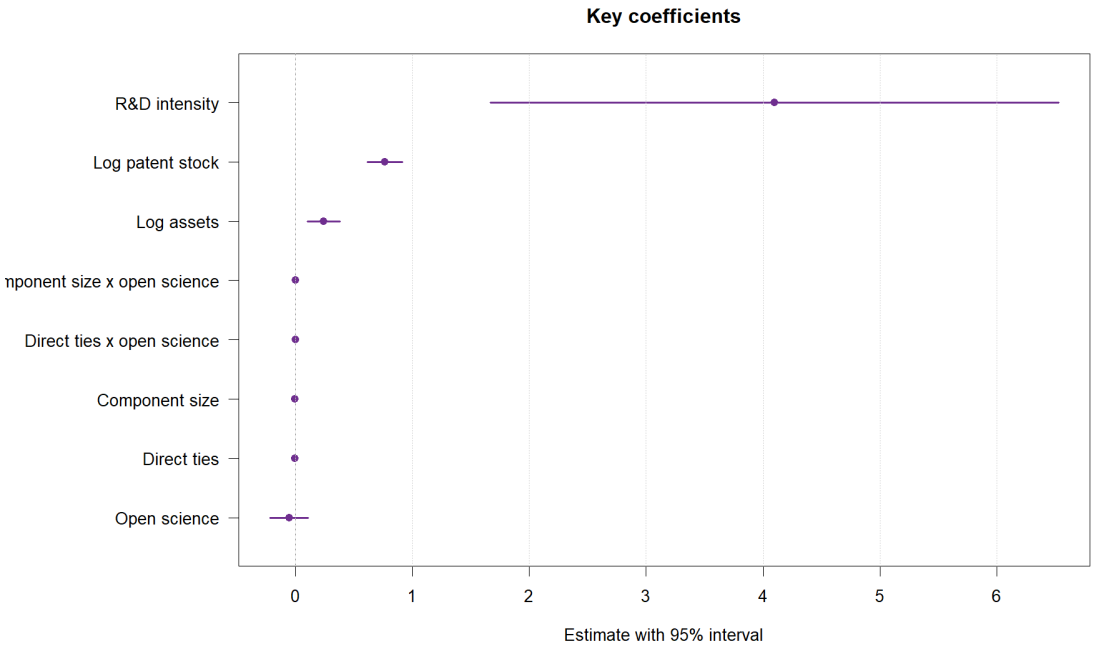

Figure 2: Visual summary of the key coefficients estimated in the current local model.

6.3 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Log patent stock is estimated at 0.764 with a p-value of ¡ .001; Log assets is estimated at 0.244 with a p-value of ¡ .001; R&D intensity is estimated at 4.097 with a p-value of ¡ .001; Component size is estimated at -0.0040 with a p-value of 0.060 (**???????**). Within the focal specification, the sharpest local signal comes from Log patent stock, which is estimated at 0.764 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (**???????**). The scale of the underlying variables also matters for interpretation. The outcome Next-year patents has mean 228.2 and standard deviation 519.3 in the local panel,

while the main regressor Open science has mean 3.440 and standard deviation 2.215 (**???????**). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 19.78 to 660.1 while the mean regressor ranges from 0.384 to 6.519. This does not substitute for the model, but it shows how much raw variation the specification must explain (**???????**). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local question is whether open-science orientation shifts the innovation benefits of network membership away from narrow centrality and toward broader component exposure. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (**???????**).

6.4 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (**????????**). Within baseline, the leading coefficient family reads as follows: Log patent stock = 0.764 (p = ¡ .001); Log assets = 0.244 (p = ¡ .001); R&D intensity = 4.097 (p = ¡ .001); Component size = -0.0040 (p = 0.060); Component size x open science = 0.00085 (p = 0.18). The largest positive estimate in this specification is R&D intensity at 4.097, while the largest negative estimate is Open science at -0.0532. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (**????????**). This fuller coefficient reading is especially important for Owen-Smith and Powell (2004) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (**????????**).

6.5 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Baseline, the most informative focal term is Log patent stock with estimate 0.764

and p = ¡ .001 (????????). The Baseline specification contributes its own reading to the replication. Log patent stock is estimated at 0.764 with p = ¡ .001 and is classified as a strong signal; Log assets is estimated at 0.244 with p = ¡ .001 and is classified as a strong signal; R&D intensity is estimated at 4.097 with p = ¡ .001 and is classified as a strong signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (????????). This model comparison matters for Owen-Smith and Powell (2004) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (????????).

Term	Model	Estimate	PValue	Signal	Reading
Direct ties	Baseline	-0.0062	0.25	Weak	Negative estimate with p = 0.25.
Component size	Baseline	-0.0040	0.060	Suggestive	Negative estimate with p = 0.060.
Open science	Baseline	-0.0532	0.52	Weak	Negative estimate with p = 0.52.
Log patent stock	Baseline	0.7644	¡ .001	Strong	Positive estimate with p = ¡ .001.
Log assets	Baseline	0.2436	¡ .001	Strong	Positive estimate with p = ¡ .001.
R&D intensity	Baseline	4.097	¡ .001	Strong	Positive estimate with p = ¡ .001.
Direct ties x open science	Baseline	-0.0007	0.38	Weak	Negative estimate with p = 0.38.

Term	Model	Estimate	PValue	Signal	Reading
Component size x open science	Baseline	0.0008	0.18	Weak	Positive estimate with p = 0.18.

Model-by-model interpretation table drawn from the current coefficient exports.

6.6 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (**???**). For knowledge_networks, the current fit export reports N = 2,276, pseudo R-squared = 0.953, R-squared = NA (**???**). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Owen-Smith and Powell (2004) is being carried out on a clearly described estimating sample with observable fit statistics (**???**).

model	n	logLik	AIC	BIC	pseudo_r2	r2	within_r2
knowledge_networks	2,276	-25,709	51,788	52,848	0.953	NA	NA

Model fit statistics discussed in the main results section.

Term	Estimate	Std. error	Statistic	P-value
Direct ties	-0.0062	0.0054	-1.159	0.246
Component size	-0.0040	0.0021	-1.878	0.0604
Open science	-0.0532	0.0826	-0.645	0.519
Log patent stock	0.7644	0.0758	10.09	6.11e-24
Log assets	0.2436	0.0714	3.413	0.0006
R&D intensity	4.097	1.240	3.304	0.0010
Direct ties x open science	-0.0007	0.0007	-0.887	0.375
Component size x open science	0.0008	0.0006	1.326	0.185

Detailed coefficient table from the current local estimation output.

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. (**?**). For Owen-Smith and Powell (2004), the local headline result is Component Size x Open Science = 0.00085 with p = 0.18. The value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than asking the reader to infer comparability from prose alone.

Aspect	OriginalPaper	LocalDataset	Reading
Main empirical result	The published paper finds that the accessibility of spillovers depends on openness and that benefits can accrue to broad network membership rather than only to centrality.	Component Size x Open Science = 0.00085 with p = 0.18.	The published paper argues that openness changes whether innovation benefits accrue narrowly to central firms or more broadly to firms embedded in a coherent network component. The local analogue translates that question into a citation-link network with a component-size measure and an open-science proxy.

Aspect	OriginalPaper	LocalDataset	Reading
Measurement and data construction	The published design constructs Boston and Boston-plus biotechnology networks, distinguishes main-component membership from betweenness centrality, and links those positions to patenting.	The local analogue uses a rolling interfirm citation-link network, a component-size measure, and an open-science indicator built from publications and public-research citations to distinguish centrality from broader network access.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target.
Sample and unit of analysis	The published sample is a set of human therapeutic biotechnology firms located in the Boston metropolitan area and observed through yearly network snapshots.	Firm-year panel with rolling citation-network measures, open-science indicators, and next-year patenting outcomes. Local sample = 2,378 observations across 177 entities.	This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone.

Comparison of the original paper's results and design to the current local dataset analysis.

7.2 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design constructs Boston and Boston-plus biotechnology networks, distinguishes main-component membership from betweenness centrality, and links those positions to patenting By contrast, the local paper relies on the following executable design: The local analogue uses a

rolling interfirm citation-link network, a component-size measure, and an open-science indicator built from publications and public-research citations to distinguish centrality from broader network access These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (**??????????**). The same caution applies to the sample itself. The published design is described as follows: The published sample is a set of human therapeutic biotechnology firms located in the Boston metropolitan area and observed through yearly network snapshots The local sample instead works from Firm-year panel with rolling citation-network measures, open-science indicators, and next-year patenting outcomes and yields 2,378 observations across 177 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (**??????????**). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The published paper argues that openness changes whether innovation benefits accrue narrowly to central firms or more broadly to firms embedded in a coherent network component. The local analogue translates that question into a citation-link network with a component-size measure and an open-science proxy. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (**??????????**). The comparison to the published paper is intentionally narrow. The published paper argues that openness changes whether innovation benefits accrue narrowly to central firms or more broadly to firms embedded in a coherent network component. The local analogue translates that question into a citation-link network with a component-size measure and an open-science proxy. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (**????????**). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. The present manuscript is classified as methodology-specific local analogue. When the original paper depends on proprietary, historical, or otherwise unavailable

preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (**????????**).

8 Discussion

The published paper argues that openness changes whether innovation benefits accrue narrowly to central firms or more broadly to firms embedded in a coherent network component. The local analogue translates that question into a citation-link network with a component-size measure and an open-science proxy (**????????**). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (**??????**). The first limitation concerns measurement. The published paper studies the Boston biotechnology community and includes richer geography and organizational-form structure (**??????**). The second limitation concerns sample translation. The local rerun uses a broad public-firm citation-link network and an open-science proxy in place of the original regional network design (**??????**). The third limitation concerns interpretation. The paper is therefore strongest as a transportability test of the channels-versus-conduits logic (**??????**).

9 Conclusion

The paper's contribution is a transparent openness-by-network-position analogue that can be estimated from the current citation-network and science panels (**??????????**). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (**??????**).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	2,378	rows
Unique Firms	177	unique_firms
Start Year	1991	year_min
End Year	2014	year_max
Mean Next Patents	228	mean_next_patents
Mean Direct Ties	37.08	mean_direct_ties
Mean Component Size	98.04	mean_component_size
Mean Open Science	3.440	mean_open_science

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Next-year patents	2,378	228.2	519.3	20.00	78.50	226.4	0	7662
Direct ties	2,378	37.08	22.75	18.00	39.00	53.00	0	99
Component size	2,378	98.04	27.99	81.00	99.00	122.0	1	123
Open science	2,378	3.440	2.215	1.609	3.367	5.329	0	9
Log patent stock	2,378	5.803	2.097	4.833	6.254	7.236	0	10
Log assets	2,378	9.388	1.341	8.480	9.310	10.26	4	14
R&D intensity	2,378	0.0417	0.0403	0.0134	0.0281	0.0594	0	1
External public-research citations	2,378	47.66	177.7	0.0000	2.000	20.00	0	2721

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Next-year patents	Direct ties	Component size	Open science	Log patent stock	Log assets	R&D intensity	External public-research citations
Next-year patents	1	0.419	-0.064	0.480	0.476	0.436	0.113	0.814
Direct ties	0	1.000	0.227	0.653	0.774	0.567	0.054	0.242
Component size	0	0.227	1.000	0.089	0.263	-0.148	0.037	-0.100
Open science	0	0.653	0.089	1.000	0.757	0.586	0.381	0.434
Log patent stock	0	0.774	0.263	0.757	1.000	0.536	0.158	0.323
Log assets	0	0.567	-0.148	0.586	0.536	1.000	-0.138	0.307
R&D intensity	0	0.054	0.037	0.381	0.158	-0.138	1.000	0.139
External public-research citations	1	0.242	-0.100	0.434	0.323	0.307	0.139	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	FirmYears	MeanNextPatents	MeanDirectTies	MeanComponentSize	MeanOpenScience
1991	132	118	23.27	111.9	2.817
1992	130	122	28.52	113.6	2.972
1993	126	130	32.61	116.2	3.103
1994	123	129	35.21	112.3	3.332
1995	122	140	37.60	113.2	3.351
1996	123	130	38.45	118.1	3.209
1997	120	170	39.77	120.0	3.179
1998	113	183	41.24	116.9	3.251
1999	106	195	41.42	111.9	3.310
2000	102	209	39.10	97.76	3.422
2001	100	219	38.85	93.15	3.435
2002	98	226	38.65	98.00	3.464
2003	100	216	39.50	94.12	3.412
2004	98	195	40.84	94.04	3.392
2005	93	250	41.40	92.98	3.425
2006	89	232	41.79	88.93	3.674
2007	83	258	42.29	83.89	3.892
2008	78	292	40.42	77.92	3.878
2009	75	387	37.21	73.96	3.884
2010	75	374	36.07	69.16	3.840
2011	72	425	36.21	71.00	3.876
2012	70	463	36.04	66.06	3.974
2013	75	468	35.12	63.56	3.927
2014	75	463	36.11	66.33	4.155

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	484	0.384	19.78
Q2	467	1.985	56.04
Q3	476	3.389	111.5
Q4	475	4.952	293.9
Q5	476	6.519	660.1

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value
direct_ties	-0.0062	0.0054	-1.159	0.25
component_size	-0.0040	0.0021	-1.878	0.060
open_science	-0.0532	0.0826	-0.645	0.52
log_pat_stock	0.7644	0.0758	10.09	¡ .001
log_assets	0.2436	0.0714	3.413	¡ .001
rd_intensity	4.097	1.240	3.304	¡ .001
direct_ties:open_science	-0.0007	0.0007	-0.887	0.38
component_size:open_science	0.0008	0.0006	1.326	0.18

Full coefficient dump exported from the local estimation script.

G Appendix G: Coverage Tables

Variable	Missing	MissingShare
Next-year patents	0	0.00%
Direct ties	0	0.00%
Component size	0	0.00%
Open science	0	0.00%
Log patent stock	0	0.00%
Log assets	0	0.00%
R&D intensity	0	0.00%
External public-research citations	0	0.00%

Missingness audit for the variables summarized in the manuscript.

Entity	Label	Observations	FirstPeriod	LastPeriod
10499	TEXAS INSTRUMENTS INC	24	1991	2014
10519	TEXTRON INC	24	1991	2014
1078	ABBOTT LABORATORIES	24	1991	2014
10983	RTX CORP	24	1991	2014
11465	WHIRLPOOL CORP	24	1991	2014
12142	ORACLE CORP	24	1991	2014
1300	HONEYWELL INTERNATIONAL INC	24	1991	2014
1356	ALCOA INC	24	1991	2014
1690	APPLE INC	24	1991	2014
1722	ARCHER-DANIELS-MIDLAND CO	24	1991	2014
2285	BOEING CO	24	1991	2014

Entity	Label	Observations	FirstPeriod	LastPeriod
2403	BRISTOL-MYERS SQUIBB CO	24	1991	2014
2663	CAMPBELL'S CO (THE)	24	1991	2014
2817	CATERPILLAR INC	24	1991	2014
2991	CHEVRON CORP	24	1991	2014
3170	COLGATE-PALMOLIVE CO	24	1991	2014
3310	CA INC	24	1991	2014
3532	CORNING INC	24	1991	2014
3835	DEERE & CO	24	1991	2014
4058	DOVER CORP	24	1991	2014

Top entities by local observation count.

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